Entanglement Beyond Bell States

What can we actually do with entanglement?

EL4043 – Teknologi dan Komputasi Kuantum

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A short discussion and preview of future topics

Today's Question

We already know how to create entanglement.

But why do we care about it?

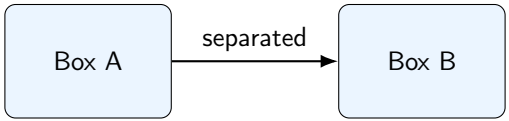
Today's goal

We will not learn a large amount of new mathematics.
Instead, we will ask:

- What makes entanglement different from classical correlation?
- Can entanglement send information?
- What can we do with entanglement?

Entanglement vs Classical Correlation

Consider two boxes.



Suppose the boxes contain correlated colors:

$$\text{A} = \text{red} \quad \Longleftrightarrow \quad \text{B} = \text{blue}.$$

If we open A and find red, we immediately know B contains blue.

Question

Is this quantum entanglement?

No. It is classical correlation.

What Makes Quantum Entanglement Different?

A Bell state is

$$\boxed{\left|\Phi^{+}\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)}$$

It cannot be written as

$$|\psi_A\rangle \otimes |\psi_B\rangle \,.$$

Therefore, the two qubits do not have independent quantum states.

Important distinction

- Classical correlation: joint behavior can be explained using pre-existing local values.
- Quantum entanglement: the joint state cannot be decomposed into independent subsystem states.

Correlation $\neq$ Entanglement

Can Entanglement Send Information?

Suppose Alice and Bob share an entangled pair.

Alice measures her qubit.

$$\left|\Phi^{+}\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Bob's result will be correlated with Alice's result.

Question

Can Alice use this correlation to send Bob a message faster than light?

The Answer: No

Entanglement produces strong correlations. But Alice cannot choose whether her measurement gives

$$0$$

or

$$1.$$

Therefore, Bob cannot determine Alice's measurement choice from his local results alone.

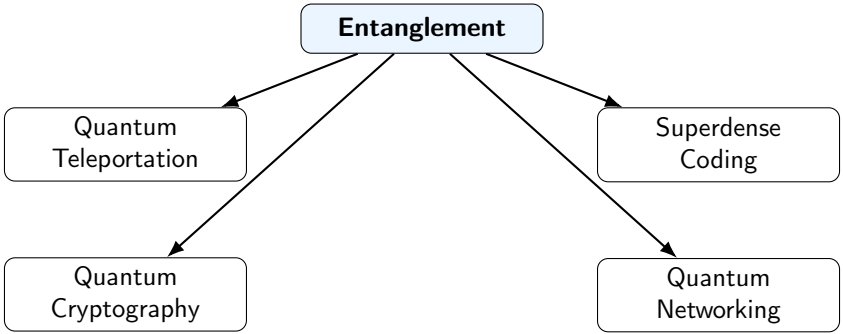
Key idea

$$\boxed{\text{Entanglement} \neq \text{Faster-than-light communication}}$$

The correlations become visible when Alice and Bob later compare their classical measurement results.

So What Is Entanglement Good For?

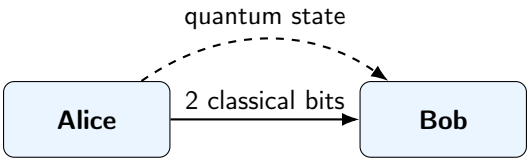
Entanglement is an important resource in quantum information.



Application 1: Quantum Teleportation

Can we transfer an unknown quantum state without physically sending the qubit?

Yes: Quantum Teleportation



The protocol uses:

Entanglement + Measurement + Classical communication

Application 2: Superdense Coding

Here is another surprising question:

How many classical bits can one qubit communicate?

Without shared entanglement:

$$1 \text{ qubit} \longrightarrow 1 \text{ classical bit}$$

With shared entanglement:

$$\boxed{1 \text{ qubit} + \text{shared entanglement} \longrightarrow 2 \text{ classical bits}}$$

Idea

Alice and Bob first share an entangled pair.
Alice then applies one of several operations to her qubit and sends that **one qubit** to Bob.

Entanglement in Real Quantum Computers

So far, our circuits are ideal. Real quantum computers are **not**.

Entanglement

↓

Interaction with environment

↓

Decoherence

Real systems experience:

- Noise & Decoherence
- Gate errors
- Measurement errors

Maintaining useful entanglement is one of the major challenges of quantum computing.

Questions for the Future

We can now ask deeper questions:

1. How can we experimentally verify that a system is entangled?
2. Can entanglement be used for secure communication?
3. How does quantum teleportation actually work?
4. Why does superdense coding work?
5. What happens to entanglement in a noisy environment?
6. How can many entangled qubits be controlled?

These are the questions we will explore later.

Take-Home Message

Entanglement is not just a strange correlation.

It is a **resource** used in quantum information.

$$\text{Entanglement} \rightarrow \begin{cases} \text{Teleportation} \\ \text{Superdense coding} \\ \text{Quantum cryptography} \\ \text{Quantum networking} \end{cases}$$

And importantly:

$$\boxed{\text{Entanglement} \neq \text{Faster-than-light communication}}$$

The interesting question is not only "What is entanglement?"

but "What can we do with it?"

Questions?

Let's see what entanglement can do for quantum computing and other quantum applications in the near future.

$$\left|\Phi^{+}\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Thank you.